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Introduction

Thank you for contributing to our open source flapping-wing vehicle project, the Cooper Union Bird (CUBird)! We appreciate your time and feedback.

The goal of this project is to create a low-cost (under \$200), accessible flapping-wing vehicle. The vehicle can be created with only a 3D printer or laser cutter, hand tools, and some off-the-shelf parts. We hope to lower the bar of entry to this research space, in addition to facilitating cheaper and simpler swarm capabilities.

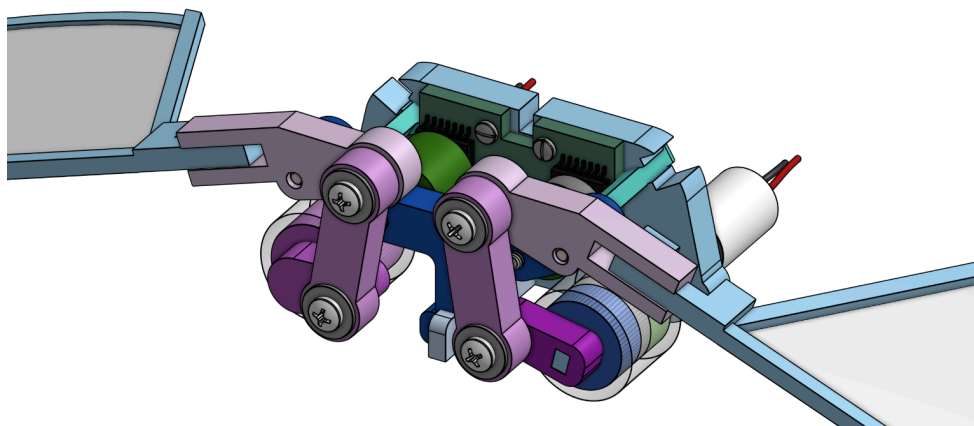
This project is currently in a prototyping state, with only the transmission and flapping control electronics presented. The design has not yet been optimized for weight or efficiency. Future work will be to complete the chassis and mounted power and control electronics.

This guide provides a step-by-step approach to manufacturing and assembling the linkages, wings, and electronics of the transmission of the CUBird. Each section is divided into parts, preparation, and build for ease of assembly. Numbered visuals are provided of each step along with more detailed explanation and helpful tips underneath the bullet points.

A parametrically designed OnShape CAD file is also provided. Parameters can be adjusted in the Geometric and Tolerancing Parameters Variable Studio tab. Tolerances are included and can be easily adjusted to fit your fabrication technique and equipment.

The Onshape file to copy can be found here:

<https://cad.onshape.com/documents/c00f7f709d6dc65e805b0aa4/w/ba05673872c8c679d2146301/e/1f776d55aba411748dcdbb73?renderMode=0&leftPanel=false&uiState=6abae341001e54ca7a42aad>



Section 1: Transmission Assembly Instructions

Parts and Materials:

Cut from 3mm thick acrylic:

- [1] Main motor base
- [2] Y-shaped front support
- [3] Magnet holder (x2)
- [4] Motor roller front (x2)
- [5] Motor roller back (x2)
- [6] Input link (x2)
- [7] Coupler link (x2)

Cut from 1.5mm thick acrylic or 3D printed:

- [8] Motor roller center
- [9] Acrylic tab (x2)
- [10] Bottom support
- [11] Side supports (x2)

3D printed components

- [12] Output link (x2, one left, one right)
- [13] Wing frame (x2, one left, one right)

- All above parts can be 3D printed instead of laser cut if desired, but tolerances will need to be adjusted according to your printer (values for our printer are included as a starting point)
 - [Laser Cut Version .dxf and .stl files](#)
 - [3D Printed Version .stl files:](#)
 - If choosing to 3D print components, [4], [5], [6], [8], and [9] can be printed as a single part (included in .stl files)

Additional components

- [14] Dual encoder board
- [15] NeFeB Dimetric magnetized magnets (x2) [6 mm OD, 2.5 mm long]
- [16] Silicone caps, 7mm long (x2) [cut from original with dimensions 5mm ID, 10 mm OD]
- [17] Delrin bushings, 3mm long (x6) (cut from bulk Delrin tubing with 1/16" ID, 1/8" OD)
- [18] DC gear motors 136:1 (x2)

Hardware and fasteners:

- [19] M1 screws (x2)
- [20] Nylon Pan Head 0-80 screws, 1/4" long (x8)
- [21] Screw size 0 Delrin washers (x10?)

[22] Nylon round spacer, 2.03 mm long (x2) [1.6mm ID, 4mm OD] Additional materials: • Super glue • 26AWG Motor wire • Passivated 18-8 Stainless Steel Pan Head Phillips Screw 0-80 Thread, 1/2" Long (not included in final build, but helpful for sizing parts and/or tapping holes) • Plastic wrap (or other thin, lightweight membrane) • Tape
<u>Assembly Tools</u>
Wire cutters (or other tool to cut Delrin tubes) Screw driver for 0-80 pan head Small needle-nosed pliers Scissors Xacto blade Needle file (and/or sandpaper) Size 0-80 tap (optional)

Printing/Cutting Notes

All .stl files can be downloaded from github. Tolerances are adjusted between laser cut and 3D printing, but will likely vary based on your equipment. A test cut/print is recommended for hole sizing.

The OnShape CAD tolerances included in the VariableStudio are for laser cutting. Most tolerances are between -0.1 mm and -0.2 mm to account for kerf. For 3D printing, replace all tolerances (listed as negative for laser cutting) to 0.2mm as a starting point, or use known tolerances for your printer. Also set "RollerExtraTolerance" to 0mm.

Recommended 3D printer settings (based on our Creality Ender 3 S1 Pro): 15% gyroid infill, include brim and supports

Preparation Steps

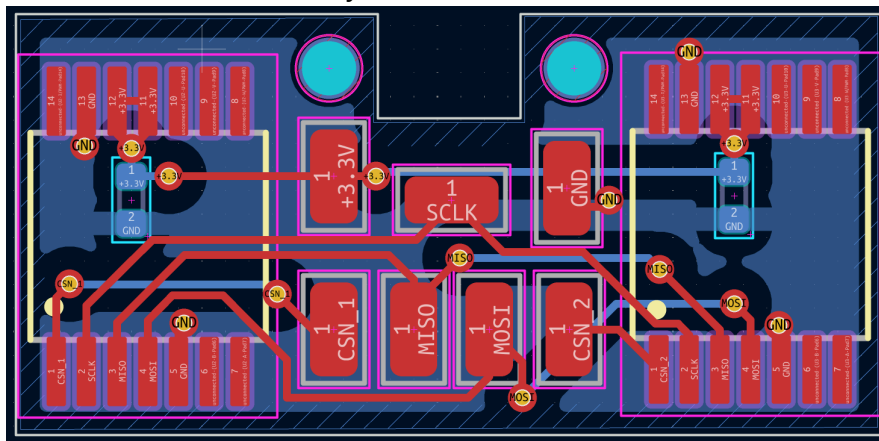
1. Cut bulk Delrin tubing into ~3 mm long sections using wire cutters or other cutter (x 6) to create Delrin bushings [17]



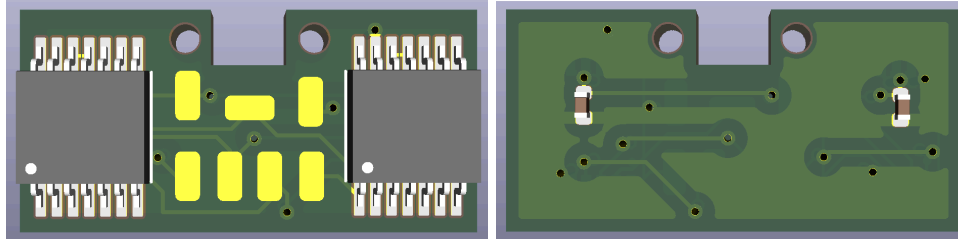
- To get a clean break and avoid clamped edges, use sharp cutters and cut quickly.
- If the bushing is deformed after cutting, insert a 0-80 screw (steel is easier) into the bearing and use needle nose pliers to gently reshape
- The Delrin is sanded flush to each bearing hole, so any length beyond 3 mm will need to be hand filed during assembly.

2. Follow instructions in Section 2 to prepare the encoder board.
 - Populate board with encoder chips and capacitors as shown in Section 2.
 - Solder pieces or 26AWG or smaller motor wire to the pads on the motor encoder breakout board [14]. Leave enough length to reach the control circuit, but not so long as to decrease signal strength (6" max).

PCB Layout with Pads Labeled



Front and back of populated board

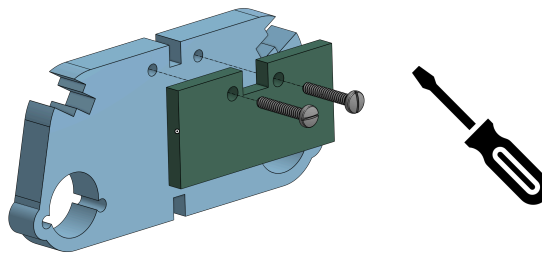


Note: See Section 2: Circuit Preparation Steps for instructions

Note: If using 3D printed parts, you may find it easier in some steps to use a small heat gun to soften the parts for assembly. Be cautious not to deform the parts

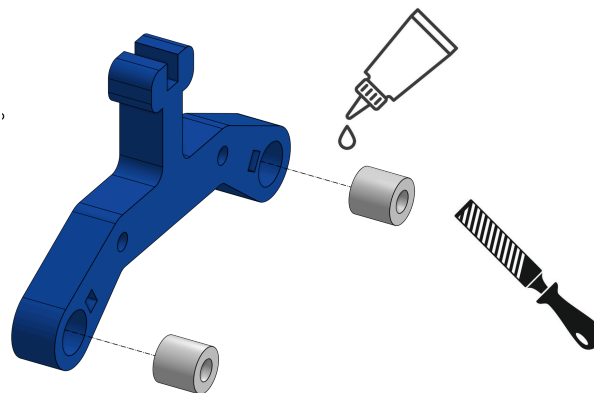
Build Instructions:

1. Attach the pre-soldered encoder board [14] into the main motor mount base [1] using two M1x6 screws [20]. The screws should self-thread into the motor mount base.

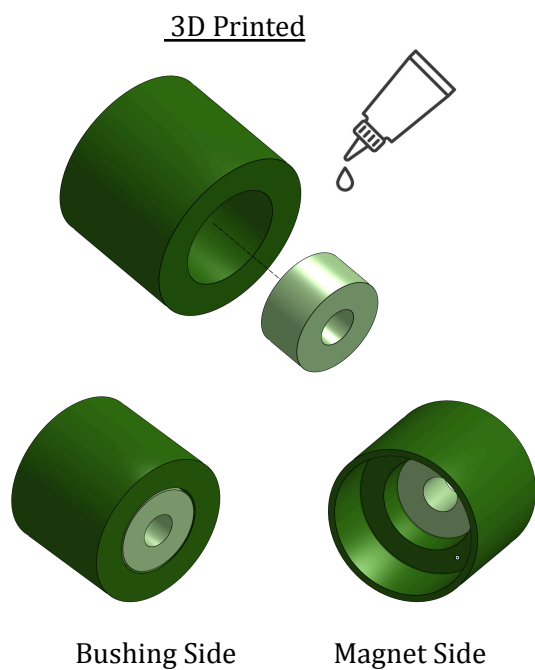
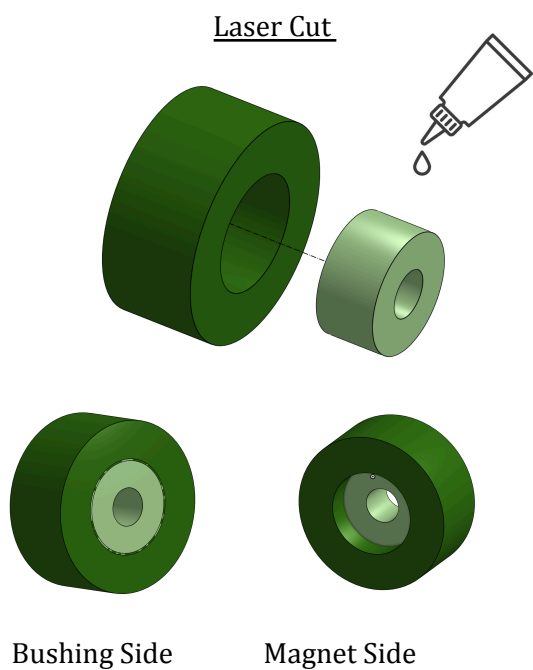


- Leave encoder wires freely hanging behind the mount, or secure with tape until Section 2, Circuit Assembly.

2. Press fit a delrin bushing [17] into each of the holes on the end of the arms of the Y-shaped front [2].
 - If desired, glue into place
 - If bushing length overhangs the width of the Y-shaped front, file until the edges are flush.



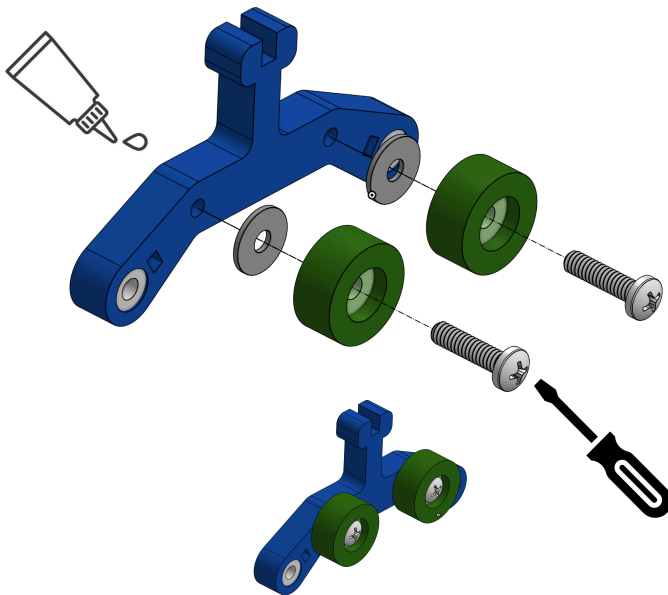
3. Prepare two magnet holder assemblies (x2).
- a. Insert Nylon spacer [22] into acrylic or 3D printed holder [3], flush with the surface of the magnet holder. Fasten with a small amount of super glue if necessary



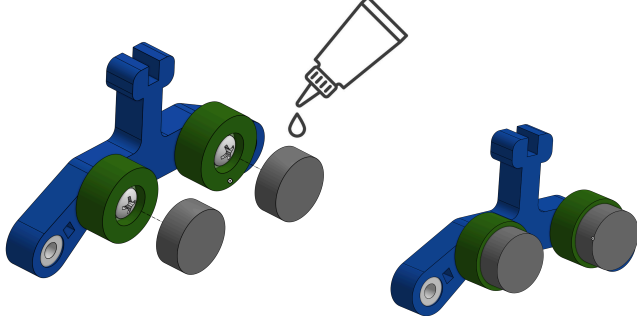
4. Attach magnet holder assemblies (step 3) to Y-shaped front (step 2).
 - a. Insert Nylon 0-80 screw [20] through the front of the assembly and spacer, ensuring that the head sits below the other edge of the holder.
 - b. Place a washer [21] on the screw between the spacer/magnet holder and the Y-shaped support
 - c. Use a screwdriver to thread the screws into the Y-shaped front and use a drop of super glue on the far side to secure the screw. You may choose to tap the holes in the Y-shaped support for easier assembly.
 - d. Super glue one magnet onto the front of the magnet holder, taking care to keep it concentric, especially if using laser cut parts. This should cover the head of the screw.

Laser Cut

4a-c:

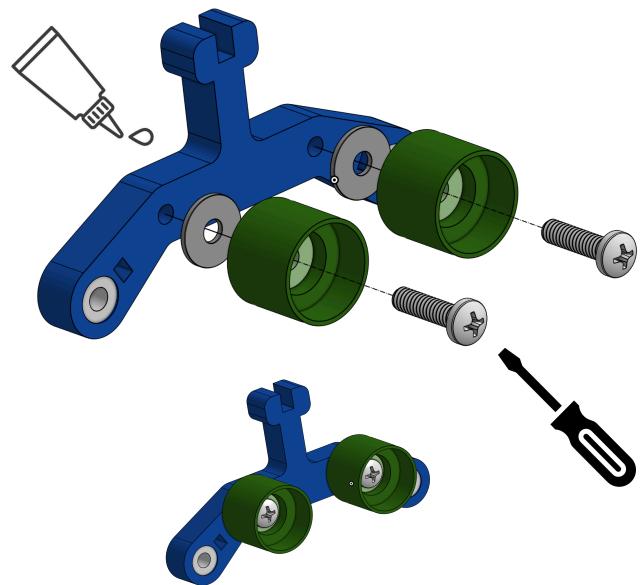


4d:

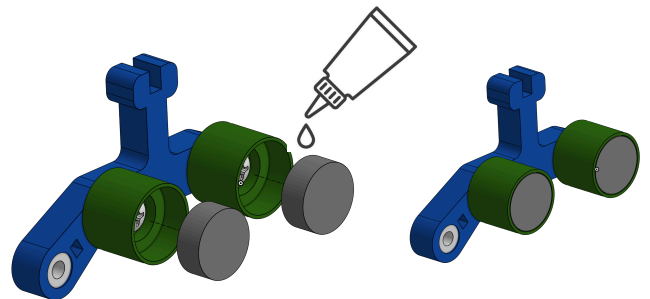


3D Printed

4a-c:



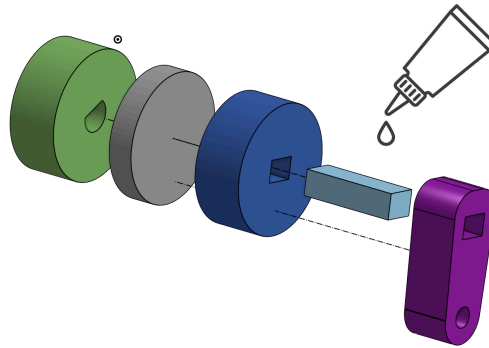
4d:



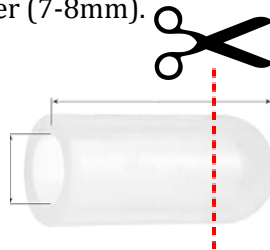
- Shafts should be perfectly orthogonal to the Y-shaped front.
- You may find it necessary in later steps to file down the distal end of the Nylon screws that protrude so they don't interfere with the linkages. This will depend on your tolerances while assembling the rest of the parts.

5. Assemble motor rollers (x2). If using the combined 3D printed part, skip to step 5d.
- Super glue the front [4], center [8], and back [5] pieces, taking care to keep them as concentric as possible.
 - Insert the acrylic tab [9] into the cutout on the front of the roller. Super glue in place.
 - Insert the input link [6] onto the acrylic tab and super glue in place

Laser Cut Only:



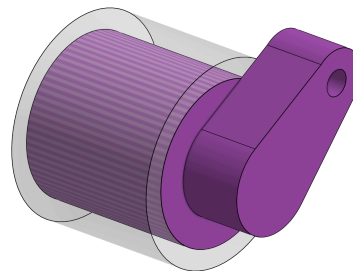
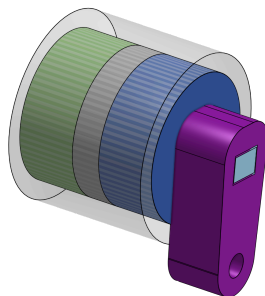
- Using scissors or a wirecutter, cut the rounded edge off the silicone cap [16] and cut to the length of the round part of the roller (7-8mm).



- Stretch and roll the silicone tube onto the rounded part of the roller.

Laser Cut

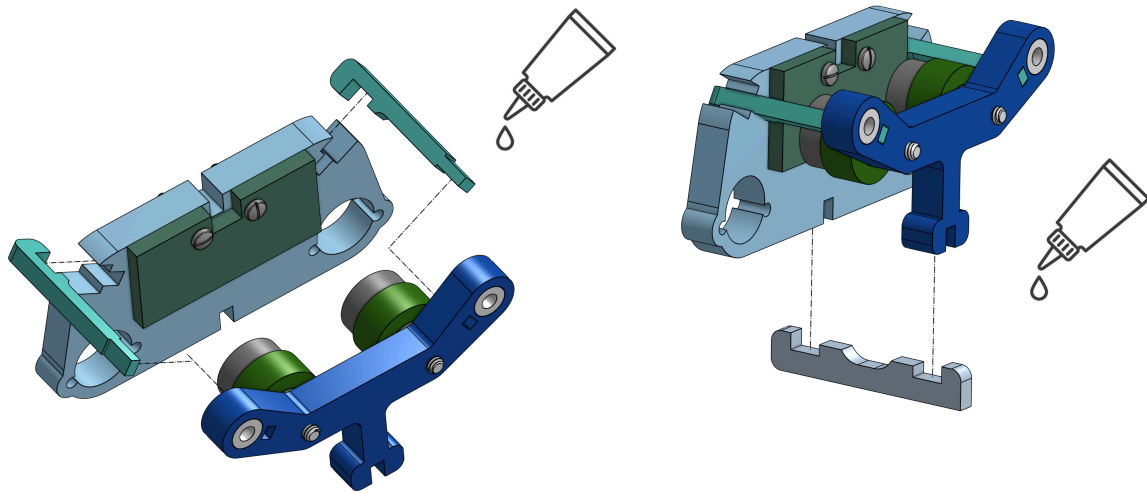
3D Printed



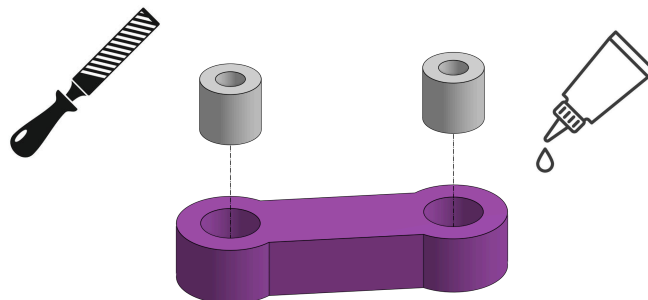
(x2)

6. Attach and glue the Y-shaped front assembly with magnet holders (Step 4) onto the main motor mount base (Step 1) using the bottom support [10] and two side supports [11]. The side supports clip into main motor base and the tab inserts into the rectangular hole on the Y-shaped front piece. The bottom

You may do this in any order that works for you! One recommendation is to align the two side supports [11] first and then the bottom support [10].



7. Prepare coupler links (x2) [7] by press-fitting Delrin bushings [17] into each hole. Glue in place if necessary. If Delrin length protrudes, hand file the bushing flush to the surface.

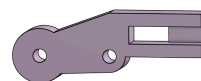
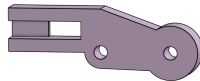


(x2)

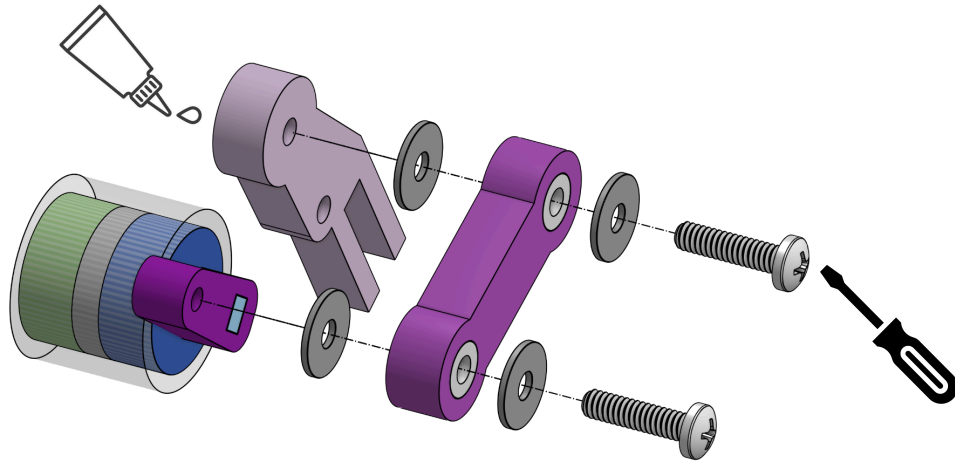
8. Assemble the four-bar linkage transmissions (x2, one left, one right).
- Insert a Nylon 0-80 screw [20] and washer [21] into one bushing on the coupler link.
 - With a washer [21] between the links, thread the Nylon screw into the input link attached to the motor roller from step 5 (or if using the 3D printed motor roller, attach directly to the integrated link). You may choose to tap the hole in the input link. Add a drop of glue to the input link end secure the screw.
 - Insert a Nylon 0-80 screw [20] and washer [21] into the other bushing on the coupler link.
 - With a washer [21] between the links, thread the Nylon screw into the designated hole on the output link [12]. You may choose to tap the hole in the output link.. Pay attention to right or left output links and their orientation. Add a drop of glue to the output link end to secure the screw.

Left

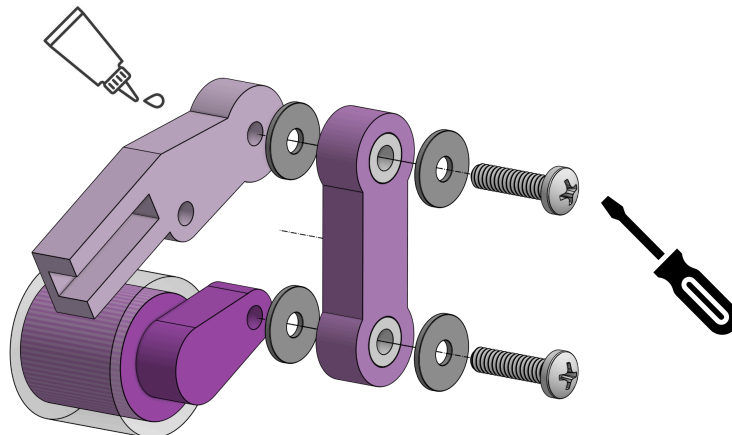
Right



Version with laser cut motor roller assembly, right side shown

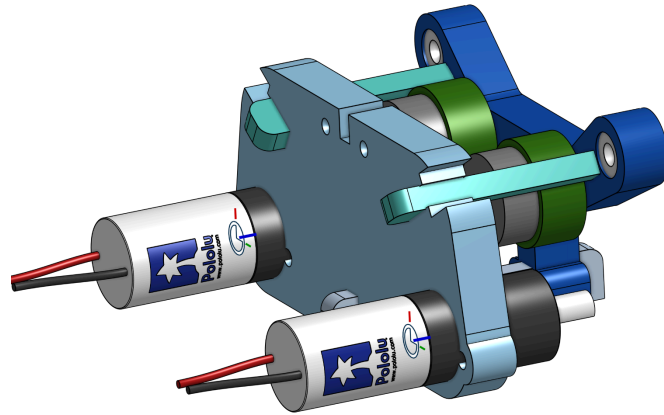


Version with 3D printed motor roller assembly, shown on left side

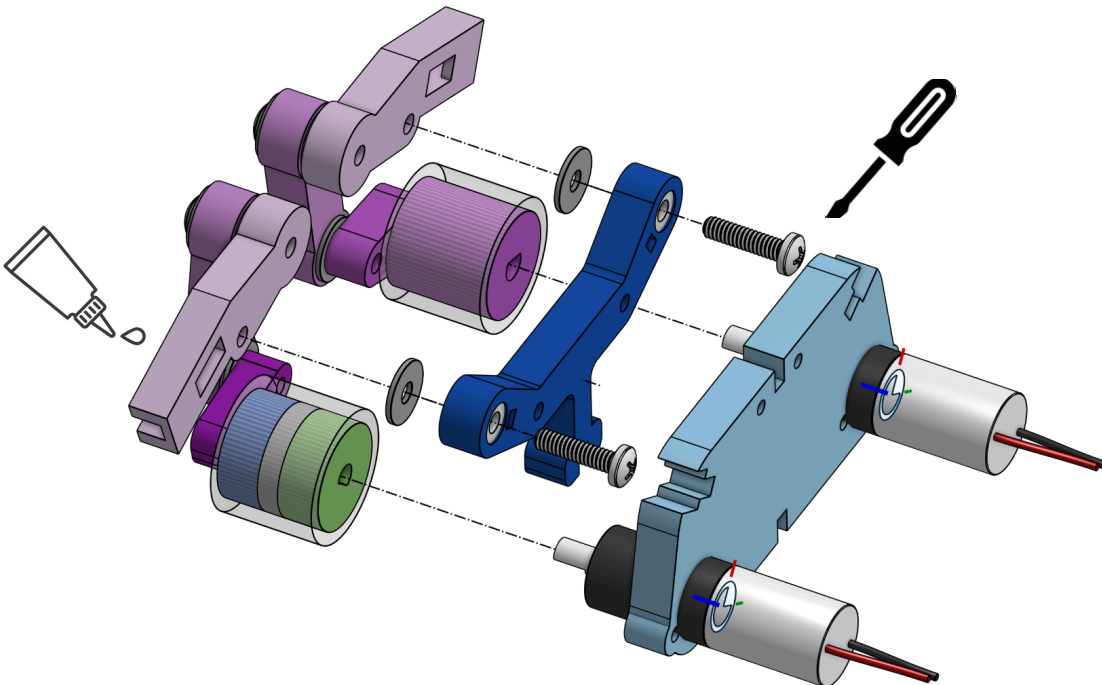


- The bushings should turn freely on the screws. If not, you can turn a longer metal 0-80 screw inside the bushing to widen it slightly.

9. Insert DC gear motors [18] through the motor mount base assembled in Step 1. At this point, they are not constrained axially, so they may shift back and forward throughout the rest of the assembly.



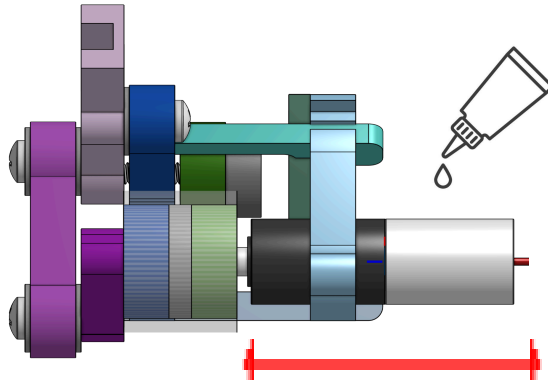
10. Attach the transmission (Step 8) to the assembled motor base (Step 9).
- Align the D-bore side of the motor roller on the prepared transmission assembly from Step 9 to the DC motor D-shaft. Pay attention to the right vs. left transmission assemblies.
 - Insert a Nylon 0-80 Screw [20] into a bushing on the Y-shaped front from the motor mount base side. With a washer between the Y-shaped piece and the link, thread the screw into the remaining hole in the output link and add a drop of glue to secure it in place.



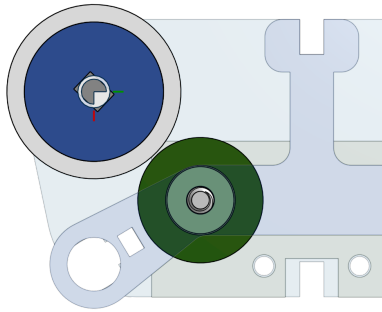
- Some already assembled parts are omitted from this view for clarity
- The axial position of the DC motor may need to be adjusted for proper alignment.

11. Ensure proper alignment and smooth motion.

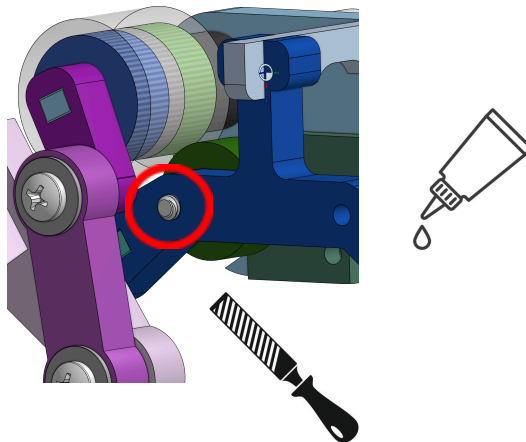
- a. The DC motor may need to be moved axially for alignment with the transmission. You may choose to add glue once it is aligned properly.



- b. The silicone on the motor roller and the magnet holder should maintain contact and turn smoothly together throughout the full rotation of the motor. If they do not, you may need to adjust the tolerances on their diameters.

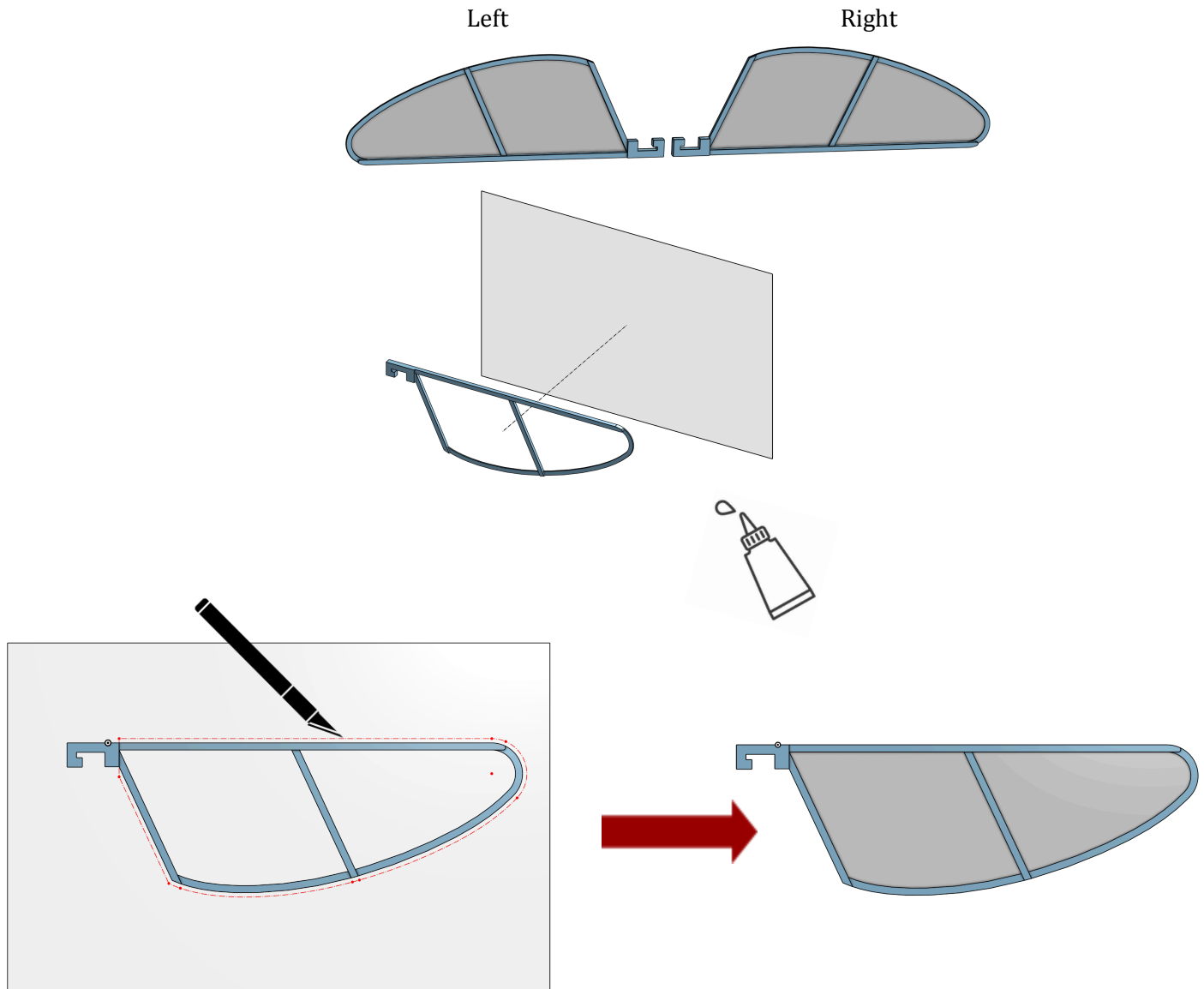


- c. If the screws used to attach the magnet holders to the Y-shaped front in Step 4 stick out too far and interfere with the transmission motion, file them down until they no longer interfere. Add another drop of glue if you filed it off to ensure that they remain in place.



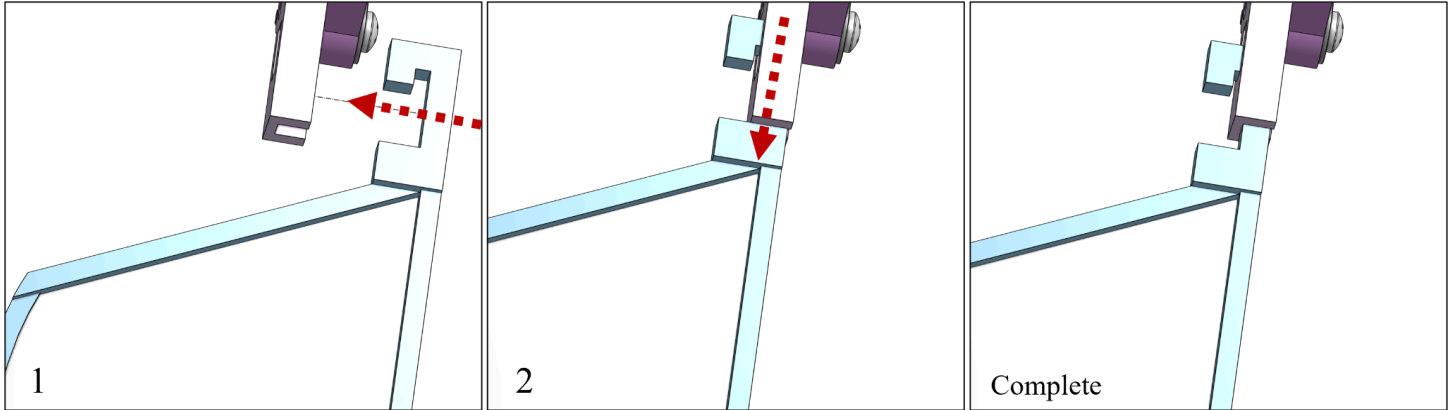
12. Assemble wings (x2 - one left, one right).

- Cut a piece of plastic wrap (or other thin membrane) and place it on a clean, flat surface. The plastic wrap should be big enough to lay the wing frame on.
- Gently smooth the plastic to remove wrinkles. Hold the plastic in place with pieces of tape, being careful to not add much tension.
- Carefully line the flat side of the wing frame [13] with super glue. The more viscous super glue may be helpful here. You may also choose to use a small piece of wire as an applicator to ensure no overflow.
- Place the wing frame glue-side down onto the smoothed plastic and press gently to ensure adhesion.
- Allow glue to fully cure.
- Release the membrane from the table and carefully cut excess material away with an Xacto blade.



13. Attach wings

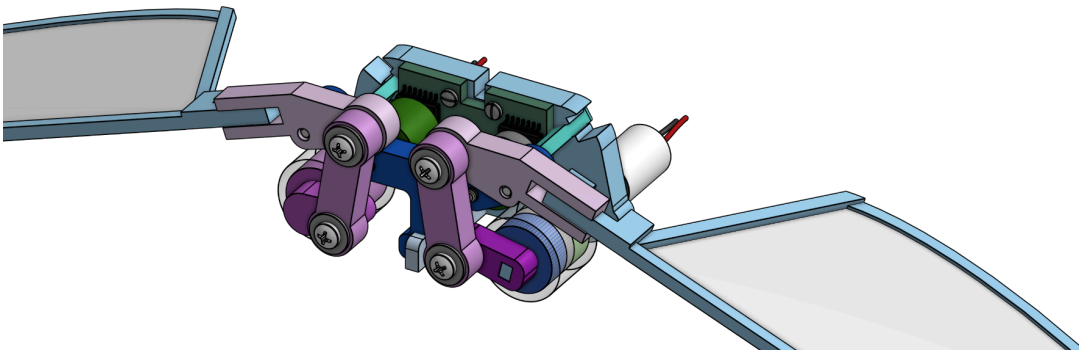
- a. Place the tab of the assembled wing from step 13 into the slot on the output link of the appropriate left or right side. The flat side (with the membrane) should face downwards
- b. Slide the wing outward to fasten in place.



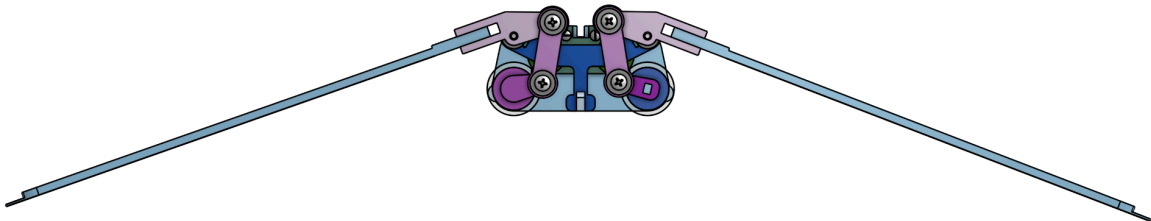
Full assembly views:

Note: The left side transmission is shown with the optional 3D printed parts and the right is laser cut.

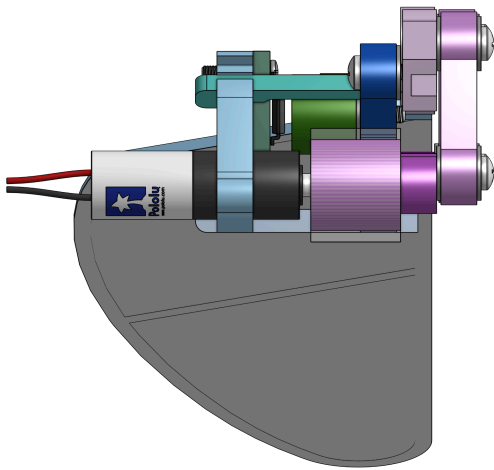
Isometric



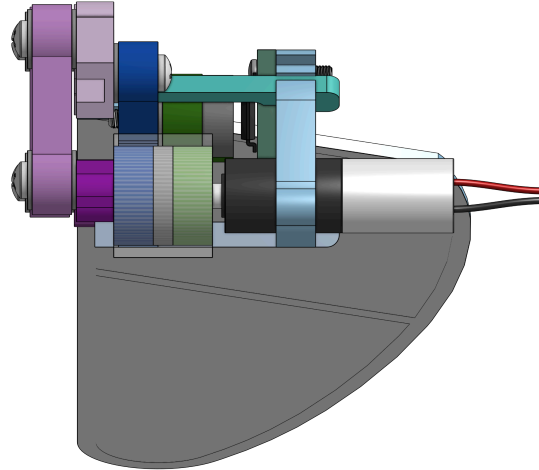
Front



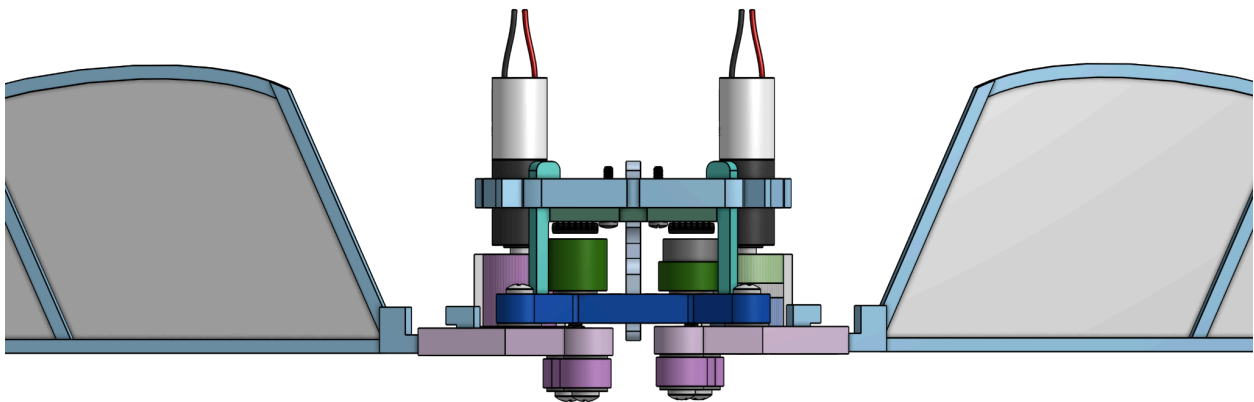
Left



Right



Top



Section 2: Electronics Assembly Instructions

The electronics assembly consists of an encoder board PCB that is mounted on the transmission (as shown in [Section 1](#)) and prototype control electronics. Buttons and LEDs are included in the control electronics section as testing and debugging tools.

Note that these electronics are in a prototyping stage and not optimized for mounting, weight, or efficiency.

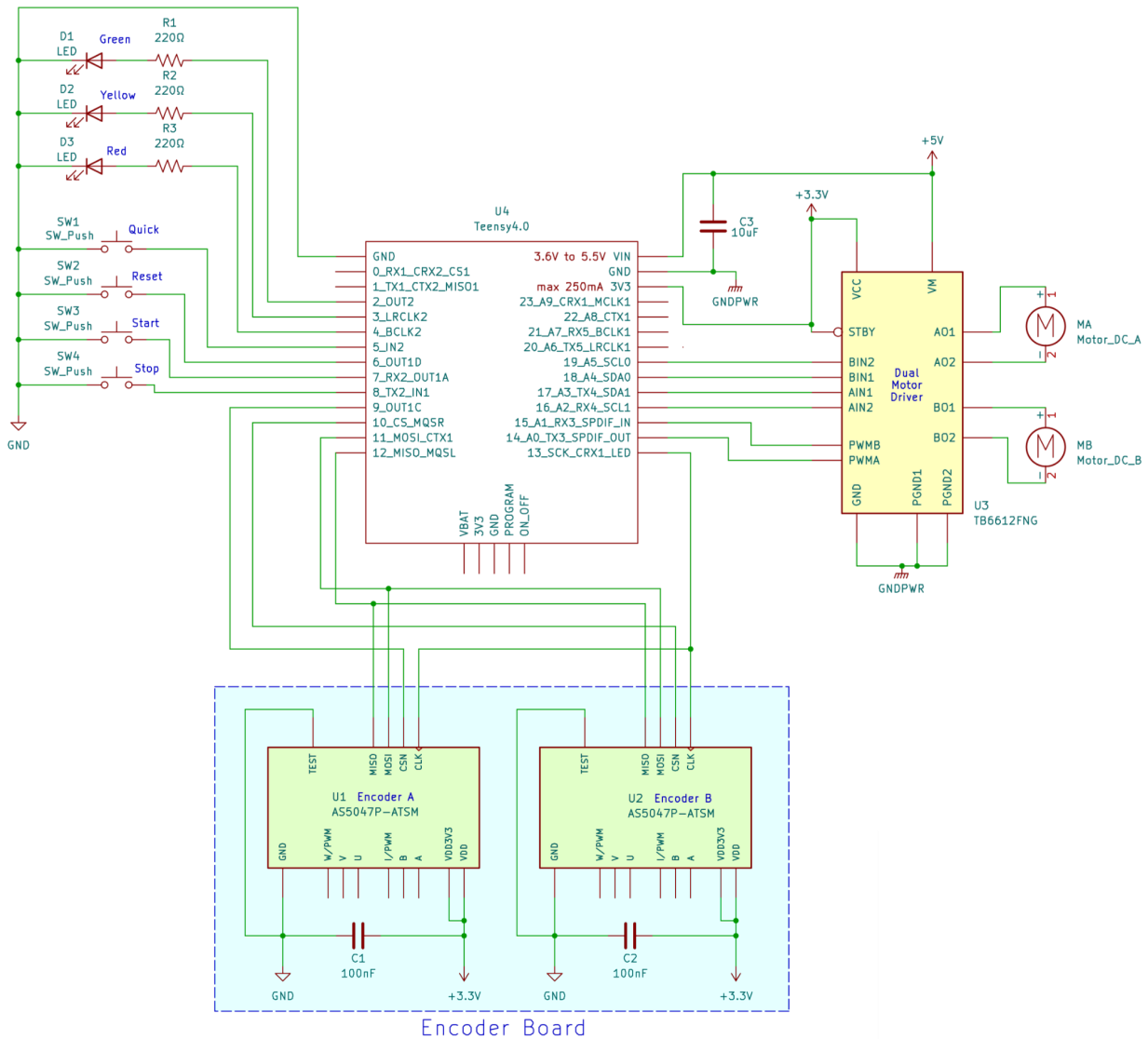
Components
Encoder Board: • Encoder Board 2-Sided PCB • Hall Effect Sensor AS5047P-ATSM (x2) • 100 nF Capacitor SMD 0402 (x2) • 24 AWG motor wire (or smaller) Control Electronics • Teensy 4.0 Development Board • TB6612FNG Dual Motor Driver • 10uF capacitor • Push button/switch (x4) • LEDs (x3, three different colors recommended) • 220 Ohm resistor (x3) • Power supply (5V and 3v3), dual breadboard power supply optional
Additional Tools and Materials:
Soldering Iron and Solder Multimeter Computer and microUSB cable (to program Teensy) Depending on how you choose to prototype the control circuit, you may want to use jumper wires, a breadboard, headers, solder, or some combination of these.

Header or Solder Selection Guide		
Prefabricated Headers (easy)	Soldered Headers (prototyping)	Soldered Wires (proper)
Jumpers	Jumpers/Solder	Wire/Solder
More expensive, but avoids soldering each pin.	Requires some soldering, but less difficulty fixing mistakes during building or reusing materials.	Most durable and secure, but creates permanent connections.

Wiring Schematic

Note: 5V and 3V3 power supplies required.

KiCAD schematics also available on github (PCB included for encoder board only)



Note: It may be convenient to put the buttons and LEDs on a breadboard.

Button/Light Control Key

STOP: turns off and clears everything (lights, motor power, flags, control history, timing, and unwrapping variables)

START: counts down with a 1 second delay between red and yellow and 0.5 second delay between yellow and green for timing

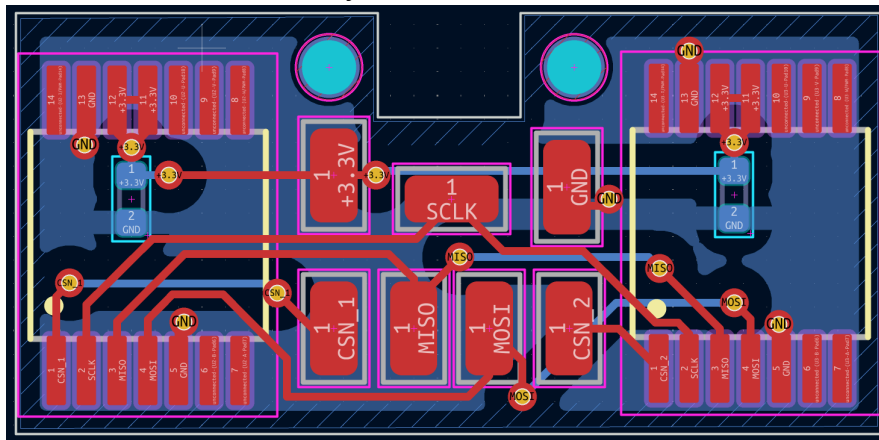
RESET: repositions the motors to absolute zero

QUICK: skips the light sequence and starts immediately

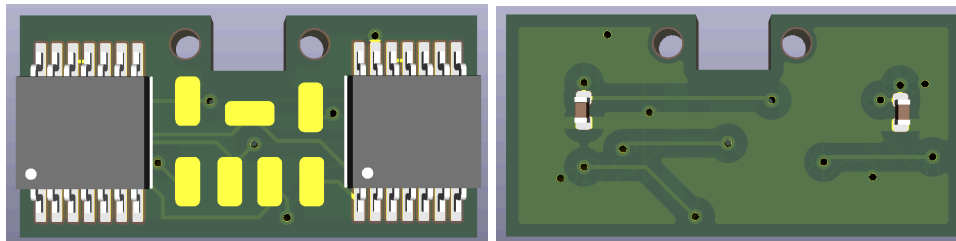
Assembly and Programming Steps:

1. Prepare the encoder board
 - a. Apply solder paste to the PCB using the solder paste stencil
 - b. Solder the Hall Effect Sensors to the appropriate pads on the board
 - c. Solder the 100nF capacitors to the appropriate pads on the board
 - d. Solder leads made from 24 AWG or smaller motor wire onto the pads.

PCB Layout with Pads Labeled



Front and back of populated board



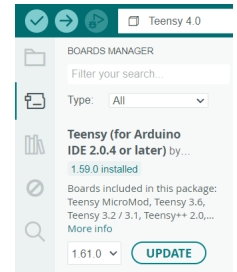
2. From the choice according to the Header or Solder Selection Guide, prepare the electrical joints:
 - If using jumpers, prepare Motor Driver and Teensy by soldering headers onto each breakout board. Otherwise, prepare solder to attach each wire according to the wiring diagram.
 - If using a breadboard, separate the rows well to avoid shorts.

Double check the connections according to the complete wiring diagram:

- Use a multimeter if possible, otherwise visually trace each connection.
- Check 3V3 versus 5V assignments to avoid overloading (could cause smoke and damage parts)
- Ensure all grounds are connected throughout the circuit.

3. Set up the programming software, used to upload the provided control code:

- Download/open Arduino IDE ([download instructions](#))
- Go to the left hand tree, open boards manager. (Or tools, board, boards manager)
- Download Teensy (including Teensy 4.0 ed.)



4. Prepare the Teensy 4.0 Microcontroller [19]:

- Attach micro-usb cable to computer port and Teensy.
- Paste provided code from github
- Upload the sketch, ensuring all serial monitors are closed.
- Select the popup Teensy window to keep visible, then press the white boot button on Teensy until the orange light flashes. Reboot OK indicates programming is done.



**Additional resources: [teensy/arduino download instructions](#) and [upload tutorial](#)*